

The Agony and the Ecstasy

Constructing a “Crash-Filtered” Equity Index using Machine Learning

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The Agony and Ecstasy

Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

Research Question

Main research question

To what extent does a “Crash-Filtered” equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between “Zombie” firms (CSI) and non-zombie firms?

Empirical design

The final panel keeps only observable CRSP firm-years: each row has the required monthly return and market-cap scaffold. Temporary and permanent CSI use the same 188,460-row scaffold; unresolved $y = \text{NA}$ rows are retained in the panel but excluded from supervised model metrics.

Data sources

CRSP

monthly returns, market cap, delistings, wealth paths

Compustat

annual accounting fundamentals

FRED

macro variables

Universe

Minimum size

market cap $\geq$ USD 100M

Observable scaffold

188,460 annual firm-year rows in both tracks

Index buckets

Total, Large, Mid, Small Cap

Sample period

Coverage

Jan 1993 – Dec 2024 (32y, monthly frequency)

Supervised labels

train/test labelled; OOS may retain unresolved labels

metrics use labelled rows only

Temporal split



Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI candidate if it satisfies all three path criteria:

- **Initial Crash (C):** more than a C drawdown from the trailing peak marks the beginning of the “zombie” period. $C \leq -80\%$
- **Non-Recovery (M):** maximum cumulative return remains below the recovery ceiling M during the zombie period. $M \leq -20\%$
- **Zombie Period (T):** duration of the confirmation window during which non-recovery must hold. $T = 18 \text{ months}$

The prediction task is annual. A trigger in calendar year $t + 1$ gives the firm-year observation at t a positive label when the track-specific confirmation rule is met:

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a qualifying CSI event within } [t, t + h], \\ 0, & \text{otherwise.} \end{cases}$$

Applying the paper classification

By strictly applying the methodology proposed by Tewari et al. (2024), one finds 8,517 response observations for the selected CRSP-dataset within the 1993 to 2024 time period. The prevalence of the response variable is similar across the Train and Test datasets. Overall, the whole dataset has 188,460 observations, with roughly 75% being clustered in the Train set.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	171,269	8,517	179,786	4.74%
Train	143,173	136,630	6,543	143,173	4.57%
Test	18,111	17,307	804	18,111	4.44%
OOS	27,176	17,332	1,170	18,502	6.32%

Motivation for robustness

Even though the paper outlines that a combination of classification grid-values were tested, the selected combination still appears arbitrary. For this reason, further robustness checks will be conducted to evaluate the chosen grid-values. It will be asked whether the dataset actually contains the bankrupt firms and how different the prevalence-rate of the response variable is when testing different grid-combinations.

Robustness I: Do CSI-firms go bankrupt?

Validation question: Among firms that eventually delist or bankrupt, how many had already been classified as CSI?

Scenario	CSI rows	CSI firms	CRSP 572-574 firms	Detected	Missed	Detection rate
Before: confirmed_csi only	8,369	5,642	629	151	478	24.0%

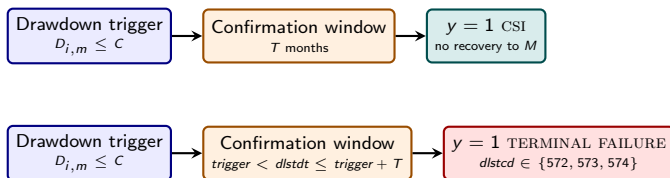
CRSP bankruptcy-related delisting codes 572, 573, and 574 are used as the terminal-distress audit set.

Core issue: The low-detection rate causes methodological issues. If one sticks strictly to the paper, then the CSI methodology should catch firms which do not recover anymore. Yet, the data shows a different result.

Methodology II: "Temporary-CSI"

Revisions to the classification methodology

The original CSI-classification grid remains unchanged. It is proposed to explicitly add all firms, which match the CRSP-codes 572,573 and 574, to the dataset after observing a valid drawdown trigger. This classification methodology will henceforth be referred to as "Temporary-CSI".



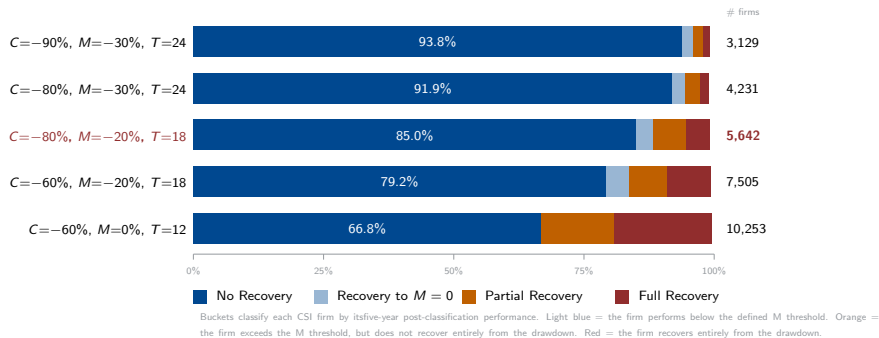
Scenario	Bankrupting	Detected	Missed	Detection
Before: CSI only	629	151	478	24.0%
After: CSI + terminal failure	629	545	84	86.65%

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

Robustness II: Do CSI-classified firms recover again?

In addition to bankruptcy checks, robustness checks were also conducted to determine whether CSI-classified firms are in persistent decline.

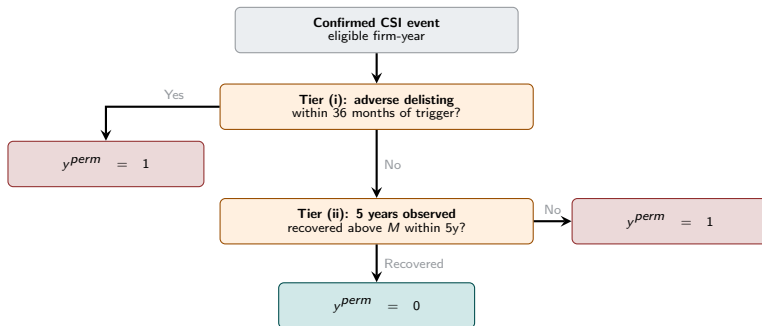
A grid-search for the three classification parameters was conducted to track the five-year recovery of firms after classification. Subsequently, each firm is sorted into one of five buckets depending on its five-year performance after classification.



Methodology III: "Permanent-CSI"

Dealing with recovering firms

To deal with the prior outlined false-positives due to observing a non-neglectible portion of recovering firms, it is proposed to test a secondary classification methodology: "Permanent-CSI". This methodology is even further forward looking, if a firm did not recover above the defined M threshold after 5 years, it will be classified as a CSI-event.



Applying the classification: Permanent CSI

By construction, Permanent-CSI is more restrictive than the initial classification methodology. The whole datasets now records 6,258 responses with $y = 1$ (compared to 8,517 originally). The prevalence drops from 4,74 to 3,34%. Overall, the prevalence rate is similar amongst the train and test set, with the test set showing a slightly smaller prevalence of the response variable.

Split	Obs. rows	$y = 0$	$y = 1$	Labelled	Prev.
Full	188,460	181,368	6,258	187,626	3.34%
Train	143,173	137,794	5,379	143,173	3.76%
Test	18,111	17,511	542	18,053	3.00%
OOS	27,176	26,063	337	26,400	1.28%

Low prevalence in the OOS set

The OOS dataset shows a markedly smaller prevalence rate of the response variable. The reason being that the forward-looking classification-methodology is constrained by the available data periods. Yet, this is not an issue as one does not require labelling the OOS dataset, which will be used to mainly evaluate the index-construction performance at the later part of the thesis.

Dataset: Descriptive Statistics

CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

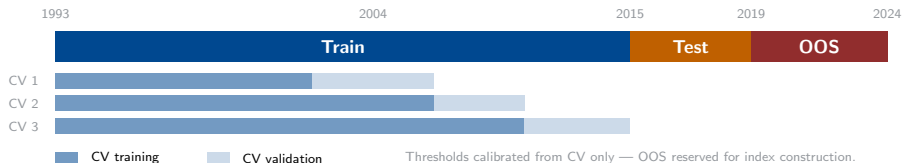
Sample	CSI	y	Obs.	Share	Firms	Median mcap	Ann. ret.	ROA
Full	Temp.	0	171,269	95.26%	19,073	254.9	4.9%	1.6%
Full	Temp.	1	8,517	4.74%	5,603	74.9	-35.0%	-18.8%
Full	Perm.	0	181,368	96.66%	19,565	252.6	4.0%	1.5%
Full	Perm.	1	6,258	3.34%	4,696	65.7	-37.5%	-20.0%

Remark: Market cap is USD millions.

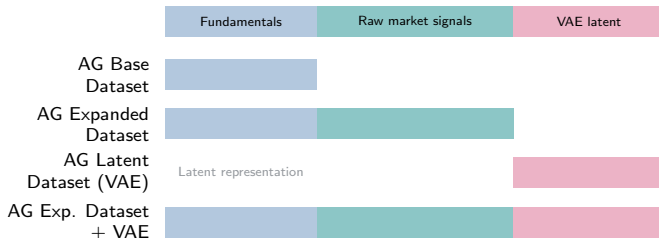
Even though the classification methodology seems to separate firms rather well, it also constrains the expectations for the index-construction part of this thesis. CSI-classified firms are considerably smaller (measured by market cap.) than their peers. Thus, their index weight will be smaller, which implies only a minor contribution to overall index-returns. Consequently, a priori one would not expect to observe a drastic outperformance by avoiding or downweighting CSI-classified firms.

Modelling I: Setup and Feature Engineering

Validation design



Final feature-set keys



Modelling II: Results Temporary CSI

Description

Temporary CSI is a rare-event classification problem: only 4.74% of labelled firm-years are positives. Because of this class imbalance, AP and fixed-FPR recall are more decision-relevant than AUC alone. AG Exp. Dataset + VAE ranks best in CV, while AG Expanded Dataset is strongest on the 2016-2019 test split.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.2046	0.8635	0.2344	0.1985	0.8766	0.2002
AG Base Dataset	0.1923	0.8446	0.2206	0.1656	0.8538	0.1517
AG Latent Dataset (VAE)	0.1659	0.8454	0.1706	0.1501	0.8438	0.1294
AG Exp. Dataset + VAE	0.2114	0.8667	0.2478	0.1864	0.8741	0.1779

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A10.

Interpretation

Test-AP improves from the 4.44% test prevalence baseline to 19.85%, showing meaningful concentration of CSI cases near the top of the risk ranking. Test-AUC of 0.8766 confirms strong broad ranking, but AP is the more useful indicator for practical screening.

Modelling III: Results Permanent-CSI

Description

Permanent CSI is the harder task: positives are rarer than temporary CSI, with only 3.34% positives in the full labelled sample and 3% in the test split. Despite this, the models still rank risk well. AG Expanded Dataset remains the reporting baseline, while AG Exp. Dataset + VAE is the strongest non-raw challenger on CV/test.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.1854	0.8757	0.2607	0.1416	0.8810	0.1808
AG Base Dataset	0.1785	0.8677	0.2379	0.1289	0.8627	0.1697
AG Latent Dataset (VAE)	0.1426	0.8460	0.1960	0.1115	0.8496	0.1513
AG Exp. Dataset + VAE	0.1883	0.8719	0.2578	0.1415	0.8838	0.2011

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A11.

Interpretation

Temporary CSI has more positives and higher test prevalence: 4.44% vs 3%. Permanent CSI is therefore harder and more affected by class imbalance. AUC is similar or slightly higher for permanent CSI, but AP is lower because the base prevalence is lower. AG Expanded Dataset and AG Exp. Dataset + VAE are nearly tied on Test-AP: 0.1416 vs 0.1415. AG Exp. Dataset + VAE has the best Test-AUC and Test-FPR3, so it is a strong challenger, but not a clean replacement for the reporting baseline.

Modelling Summary: the Models Rank Implosion Risk Well

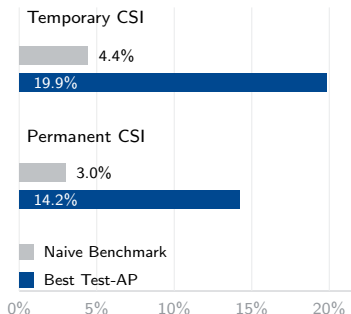
AutoGluon ensemble

Expanding-
window CV

4 feature sets

AP & recall@FPR

Test AP vs naive benchmark (prevalence)



Result: $AP \approx 4.5\text{--}4.7\times$ the base rate;
 $\text{Test-AUC} \approx 0.88$; $\text{recall@FPR3} \approx 0.20$.

Interpretation: AG Expanded Dataset is the reporting baseline; AG Exp. Dataset + VAE the strongest challenger.

Subfinding: VAE augments the base models, but it is not able to replace them.

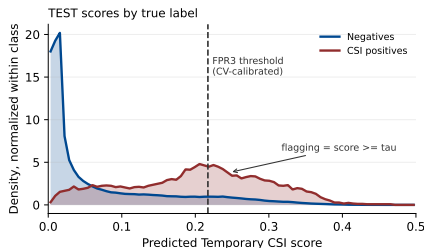
Take-Away

Calibrated scores become the index exclusion signal. The question expands from "Can we rank risk?" to "Does removing flagged firms using the model-predictions improve the risk-return metric of risk-filtered indices?"

Remark: Best Test-AP: temporary 0.199, permanent 0.142. Metrics in Appendix A10 to A13.

Why the Screen Cannot Be Perfect

The shown example is temporary CSI, AG Expanded Dataset. The exclusion threshold is calibrated on the training set to ensure for no data leakage, then carried *unchanged* to the held-out 2016–2019 test set.



On the test set at this τ

- 369 of 804 CSI firms flagged; 1,502 false positives.
- Only **10.8%** of FPs sit just above τ (within $[\tau, \tau + 0.01]$).

Conclusions for index construction

Many false positives overlap the true-positive scores. A higher threshold (e.g. with FPR5) cannot remove them without dropping real CSI firms.

Remark: CV: 72,223 rows, 3,119 positives; test: 18,111 rows, 804 positives. AG Expanded Dataset, temporary CSI.

Modelling IV: The Benefits and Drawbacks of VAE

Sub research-question. How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?

Benefits

- AG Latent Dataset (VAE) is considerably smaller in dimension (25 features) compared to AG Expanded Dataset (478 features.).
- AG Latent Dataset (VAE) performs reasonably well. AUC is similar, AP is substantially lower on CV and test.
- AG Exp. Dataset + VAE leads CV-AP in both temporary and permanent CSI.

Drawbacks

- AG Latent Dataset (VAE) cannot fully replicate the informational content of AG Expanded Dataset.
- AG Expanded Dataset retains slightly higher temporary OOS AUC: 0.8961 vs. 0.8950 compared to AG Exp. Dataset + VAE.
- Added architecture and compute cost.

Summary

Autoencoder features slightly improve average precision when added to the existing dataset, but do not dominate AUC and AP on their own. Yet, their performance is impressive given the strong reduction in dimensionality.

Modelling V: Implications for index construction

MAIN CONSIDERATION

The models **rank risk well**, but false positives are **frequent and unavoidable**.



TAKE-AWAY

Index gains are **bounded** — **difficult to beat the benchmark's risk-return**.

Low precision

AUC ≈ 0.88 , but Test-AP ≈ 0.20 — about **4 healthy firms** excluded per CSI firm.

No clean cutoff

100% of false positives overlap true-positive scores; only **10.8%** are threshold-adjacent.

Threshold drift

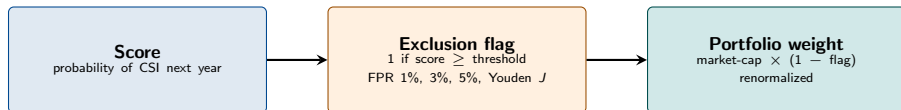
A 3% FPR cutoff on CV becomes **8.7%** on test — out-of-sample drift, not a 3% test-FPR.

Index Construction I: From Score to Portfolio Weight

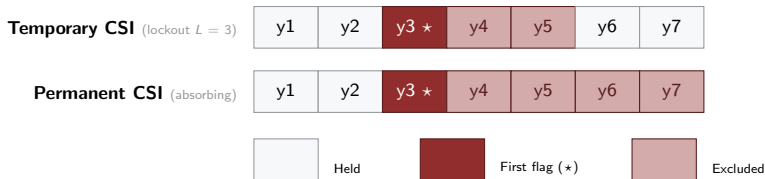
One score rule, two exclusion regimes

The model produces a probability of a CSI in the next year. A threshold turns this probability into a binary exclusion flag. The flag is then applied to market-cap weights. The two regimes differ only in how long the flag persists after it first emerges.

From model score to portfolio weight



How the flag evolves over time — a firm crosses the threshold in year 3



Index Construction II: Universes and Benchmarks

Same exclusion rule, four universes

The calibrated score-to-weight methodology is applied within each market-cap universe. Every screened index is judged against its matching market-weighted benchmark.

Total Market

all eligible firms
 $\geq$ USD 100M

Large Cap

prior-Dec. rank
top tier

Mid Cap

prior-Dec. rank
middle tier

Small Cap

prior-Dec. rank
bottom tier

Universe construction

- Prior-December market cap; minimum USD 100M.
- Large/Mid/Small split by market-cap rank.
- Quarterly rebalancing.

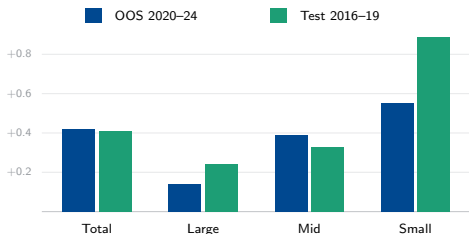
Remark: The constructed indices match the ones available via the WRDS database. To the misfortune of the author, WU does not grant access to these prepared index-data.

Benchmarks

- Primary: the market-weighted index acts as the main benchmark.
- Min-volatility Comparison: will be shown at the end of the presentation.

Index Results: Does the Crash Filter Help?

Alpha relative to the BM by universe (pp, 10 bps net)



OOS alpha. Current OOS deltas are positive in all four universes; Mid and Small are strongest.

Test-set validation. The CSI-methodology also works within the test-set. Permanent-CSI Risk/Volatility ratio rises from 0.89 → 1.00.

Summary

The crash-filter appears beneficial in both evaluation windows, but the evidence differs by split. The OOS result is a positive benchmark-relative return delta across the four universes. The test-set results are shown separately and serve as a comparison within a different market regime.

Temporary-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe net of 10 bps costs vs. the market-weighted BM.

Universe	Strategy	Return		Risk	
		Geo ret.	Δ pp	Risk/Vol.	Max DD
Total	AG Base Dataset; Youden 3y	13.3 $\rightarrow$ 13.7	+0.42	0.60 $\rightarrow$ 0.64	-25.0 $\rightarrow$ -23.8
Large	AG Base Dataset; Youden 3y	14.3 $\rightarrow$ 14.4	+0.14	0.66 $\rightarrow$ 0.68	-24.8 $\rightarrow$ -23.9
Mid	AG Base Dataset; FPR5 5y	9.7 $\rightarrow$ 10.0	+0.39	0.41 $\rightarrow$ 0.42	-24.6 $\rightarrow$ -24.5
Small	AG Exp. Dataset + VAE; Youden 3y	7.7 $\rightarrow$ 8.2	+0.55	0.31 $\rightarrow$ 0.33	-29.1 $\rightarrow$ -29.3

Main takeaway

The risk-filter works better the smaller the market cap.: the alpha rises from +0.14pp (Large) to +0.39pp (Mid) to +0.55pp (Small). Smaller-cap universes include more firms in distress and exclusions move a larger part of the portfolio (about 85-106% turnover vs. 10-13% for the total market). Drawdown improves everywhere except for small caps.

Permanent-CSI Results: OOS (2020-2024) with 10 bps of Transaction Costs

Best strategy per universe net of 10 bps costs vs. the market-weighted BM.

Universe	Strategy	Return		Risk	
		Geo ret.	Δ pp	Risk/Vol.	Max DD
Total	AG Exp. Dataset + VAE; FPR5	13.3 $\rightarrow$ 13.5	+0.26	0.60 $\rightarrow$ 0.61	-25.0 $\rightarrow$ - 24.7
Large	AG Exp. Dataset + VAE; FPR5	14.3 $\rightarrow$ 14.5	+0.22	0.66 $\rightarrow$ 0.67	-24.8 $\rightarrow$ - 24.5
Mid	AG Base Dataset; FPR5	9.7 $\rightarrow$ 10.3	+0.62	0.41 $\rightarrow$ 0.44	-24.6 $\rightarrow$ - 24.5
Small	AG Latent Dataset (VAE); FPR3	7.7 $\rightarrow$ 7.9	+0.24	0.31 $\rightarrow$ 0.32	-29.1 $\rightarrow$ -29.3

Main takeaway

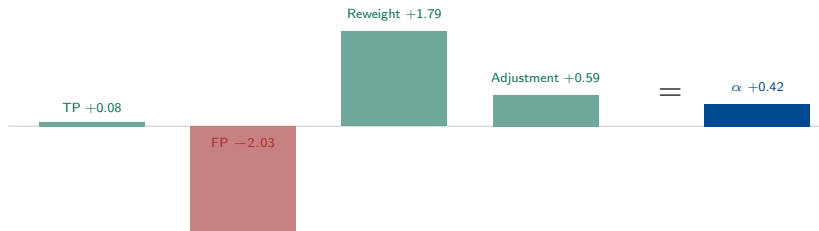
The permanent filter tilts the gains toward Mid Caps (+0.62pp), its strongest universe. Direction matches the temporary track, but more exclusion over-removes firms with broad thresholds, which is why Youden is dropped in favour of FPR3 and FPR5.

Where Does the Alpha Come From?

It is not direct event-avoidance

Because positives are rare, realized alpha is dominated by reweighting retained firms, not by directly avoiding CSI events. A large false-positive opportunity cost is the main offset. For this reason, most of the performance gain disappears.

Decomposition for Total Market, OOS (pp): $TP + FP + \text{reweight} + \text{cost} + \text{adjustment} = \alpha$



Universe (OOS, temp)	TP	FP	Reweight	TC	Geo	Alpha
Total	+0.08	-2.03	+1.79	-0.01	+0.59	+0.42
Large	+0.00	-1.50	+1.24	-0.01	+0.41	+0.14
Mid	-0.00	+0.05	+0.27	-0.11	+0.18	+0.39
Small	+0.42	-4.00	+3.94	-0.09	+0.28	+0.55

Permanent-CSI Results: Test (2016-2019) with 10 bps of Transaction Costs

Universe	Strategy	Return		Risk	
		Geo ret.	Δ pp	Risk/Vol.	Max DD
Total	AG Exp. Dataset + VAE; Youden	13.8 → 14.2	+0.45	0.89 → 1.00	-14.6 → -12.8
Large	AG Exp. Dataset + VAE; Youden	14.2 → 14.4	+0.26	0.95 → 1.04	-13.8 → -12.2
Mid	AG Exp. Dataset + VAE; Youden	13.7 → 14.2	+0.43	0.84 → 0.95	-15.3 → -13.8
Small	AG Exp. Dataset + VAE; Youden	11.4 → 12.2	+0.83	0.58 → 0.68	-20.1 → -18.3

Validation Summary

In the different 2016-2019 market-regime a single consistent strategy (Exp. Dataset + VAE with Youden computed thresholds) improves Risk/Volatility Ratios and cuts drawdown in every universe, with Total Risk/Vol. crossing 1.0. The same small-cap tilt appears (+0.83pp Small vs. +0.26pp Large).

Results of the key-cutoffs across the two CSI-methodologies

Temporary CSI

finite lockout · 3–5 yr



flag raised → released

A wrong flag **self-corrects** once the lockout expires.

More permissive cutoff: Youden

mean best alpha +0.24 pp

Permanent CSI

absorbing · irreversible



flag raised → stays forever

A wrong flag is **stuck for good**.

Stricter cutoff: FPR3 / FPR5

FPR +0.24/+0.22 pp · Youden −0.31 pp

The cutoff is determined by the CSI-methodology. Youden allows for more CSI-events in the temporary-CSI sample.

Remark: Mean best benchmark-relative alpha across universes at 20 bps.

Summary: Risk Screening, Not Crash Avoidance

1

Models rank implosion risk well

AUC ≈ 0.88 ; AP $\approx 4.5\times$ the base rate.

2

But the exclusion cutoff is unavoidably noisy

False positives overlap true positives: ≈ 4 healthy firms flagged per real CSI.

3

So the gain comes from reweighting, not avoidance

The false-positive cost is the main drag; retained-name reweighting carries the alpha.

4

And the effect is dilution-bounded

CSI firms are $\approx 1\%$ of cap — the gain is largest in Mid and Small Cap.

A bounded risk-reduction and portfolio-composition tool — not a crash-avoidance alpha engine.

Reliably cuts drawdown and modestly lifts

Risk/Vol.: useful, principled, and honestly limited.

Remark: All results net of 10 bps; OOS 2020–2024. Attribution: portfolio-mechanism evidence, not causal proof of avoided crash events.

Appendix: AI Disclaimer

The following AI-assisted language and coding models were used in the preparation of this work: Claude (Sonnet 4.6 and Opus 4.8, Anthropic), ChatGPT Codex (Codex 5.3 and GPT 5.5, OpenAI).

Claude and ChatGPT Codex were used to support the implementation of the coding pipeline, including code generation and debugging, validation of model outputs, summarization of results, and the production of charts and figures. ChatGPT Codex additionally served as the execution environment for running the code pipeline. Claude was also used in a minor supporting capacity during the preparation of presentation slides; however, all slides were conceived, structured, and finalized by the human author, with AI assistance representing a negligible share of the final content.

All AI-generated outputs were reviewed, critically assessed, and adapted by the author. The data handling and model analysis was performed locally. Model training and index-construction was implemented via Cloud Computing (Vast.ai and GCP).

Zaki Tewari, Michal Galas, and Philip Treleaven. Predicting catastrophic stock implosions: A machine learning approach. Technical report, University College London, 2024. Working Paper.

Appendix A1: Dataset, Labels, and Detection Checks

Failure-label audit

Signal	Firms	Detected	Rate
CRSP 572–574	629	545	86.65%
CRSP 574 only	548	513	93.61%
Adverse 400–490, 550–585	5,654	3,676	65.02%

Temporary CSI has 8,517 positive rows and 8,674 unresolved rows. Permanent CSI has 6,258 positive rows and 834 unresolved rows.

Clean labelled prevalence

Track	Split	Labelled	Pos.	Prev.
Temp.	Train	143,173	6,543	4.57%
Temp.	Test	18,111	804	4.44%
Temp.	OOS	18,502	1,170	6.32%
Perm.	Train	143,173	5,379	3.76%
Perm.	Test	18,053	542	3.00%
Perm.	OOS	26,400	337	1.28%

Read

The labels stay rare after the observable-scaffold cleanup. CRSP terminal-failure information is an additive audit/override around the CSI timing rule, not an independent default label.

Source: Sources: AE-PANEL label-count and CRSP-overlap evidence used in the original appendix source map.

Appendix A2: Descriptive Statistics and Feature Groups

Positive firms look distressed before modelling

Track	Class	Mkt cap	Ann. ret.	ROA
Temp.	y=0	350.9	4.9%	1.6%
Temp.	y=1	68.7	-35.0%	-
				18.8%
Perm.	y=0	249.3	4.2%	1.5%
Perm.	y=1	66.2	-37.5%	-
				19.3%

Market cap is the full-sample median in USD millions.

Source: Sources: descriptive-statistics CSVs and feature-dictionary evidence from the original appendix/source map.

Feature blocks retained

- Point-in-time accounting ratios: leverage, liquidity, profitability, accruals, cash flow, and Altman-style distress.
- Firm histories: differences, accelerations, expanding means/volatility, rolling minima/maxima/trends, and peak/trough moves.
- Price and macro signals: momentum, volatility, drawdowns, macro levels, and selected firm $\times$ macro interactions.
- Feature sets: base, expanded, VAE latent, and expanded plus VAE.

Appendix A3: CV Design and Model Metrics

Expanding-window CV

Fold	Training	Validation
1	Initial block	seed only
2	through 1998	1999–2004
3	through 2004	2005–2009
4	through 2009	2010–2015

Split-before-preprocessing discipline is retained: winsorization, imputation, PIT scaling, and thresholds are fitted on training/CV information only.

Payoff metrics used in the main story

Track	Split/model	AP	AUC	R@FPR3
Temp.	Test AG Exp.	0.1985	0.8766	0.2002
Temp.	OOS AG +VAE	0.3152	0.8950	0.2632
Perm.	Test AG Exp.	0.1416	0.8810	0.1808
Perm.	Test AG +VAE	0.1415	0.8838	0.2011
Perm.	OOS VAE latent	0.0482	0.8337	0.1484

Metric interpretation

AP and fixed-FPR recall are emphasized because positives are rare. Economic claims are made only after thresholding, exclusions, turnover, and returns in the index section.

Source: Sources: 03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv and raw derived metrics.

Appendix A4: VAE and Model-Family Evidence

VAE setup

Item	Setting
Encoder	$n \rightarrow 256 \rightarrow 128 \rightarrow 64$
Latent dim.	24
Decoder	$z \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow n$
Loss	Recon. + β KL + γ CSI loss
Weights	$\beta = 1.0, \gamma = 0.1$

Test and OOS labels are not used for VAE fitting.

Source: Sources: model-suite family-winner CSVs, compact hyperparameter JSONs, ensemble-weight JSONs, and VAE evidence files.

Model-family result

- All eight final track $\times$ feature-set AutoGluon leaderboards select `WeightedEnsemble_L2`.
- Useful base signal is mostly tree/boosted-tree based; neural nets were tried but do not dominate.
- AG Expanded is the reporting baseline; AG Expanded + VAE is the strongest challenger in the main modelling story.
- VAE augments the expanded predictors; it does not replace tabular fundamentals.

Appendix A5: Index Grid and Selected Rules

Completed economic grid

- Tracks: temporary CSI and permanent CSI.
- Universes: Total, Large, Mid, and Small.
- Models: AG Expanded, AG Base, AG Latent (VAE), and AG Expanded + VAE.
- Thresholds: Youden, FPR1, FPR3, and FPR5.
- Transaction costs: 0, 5, 10, and 20 bps on strategy turnover.

Rules used in the final selected strategies

- Temporary CSI uses finite exclusion lockouts: 1, 2, 3, or 5 years.
- Permanent CSI is absorbing after the first active signal.
- Temporary winners mostly use Youden with longer finite lockouts.
- Permanent winners mostly use FPR3/FPR5, where the absorbing rule makes false positives more costly.

Interpretation

The rules translate calibrated risk ranks into implementable exclusions. Model quality is necessary, but realized alpha also depends on event timing, universe weights, turnover, and benchmark returns.

Source: Sources: `full_grid_manifest.csv`, `best_by_track_index_cost.csv`, threshold-family summaries, and turnover summaries.

Appendix A6: Transaction Costs and Attribution Sources

20 bps robustness summary

Track	Univ.	Alpha 0	Alpha 20
Temp.	Total	+0.43	+0.41
Temp.	Large	+0.15	+0.12
Temp.	Mid	+0.51	+0.28
Temp.	Small	+0.64	+0.45
Perm.	Total	+0.27	+0.25
Perm.	Large	+0.23	+0.20
Perm.	Mid	+0.74	+0.51
Perm.	Small	+0.32	+0.16

Attribution/source paths

- Headline winners and TC overlays come from AE-INDEX-SUITE final tables and comparison outputs.
- Contribution slides use config-level attribution plus model-specific error-cost decompositions.
- Test-set contribution rows use main-suite period=test attribution where isolated test diagnostics are unavailable.
- Active alpha is measured versus the zero-cost market-cap benchmark; only the strategy pays transaction costs.

Source: Sources: 03_Data_Output/7_IndexConstructionValidation/nonraw_index_suite/ comparison/final-table outputs and AE-ATTRIB evidence.

Appendix A7: Temporary CSI Sensitivity Detail

Role	Run ID	Pos.	OOS AP	OOS AUC	11C alpha	Status
Composite	C090_M000.T012	12,761	0.557	0.920	+0.169	complete
AP winner	C060_M000.T012	31,596	0.566	0.864	+0.052	reused
11C winner	C090_M020.T018	6,035	0.274	0.911	+0.226	complete
Baseline	C080_M020.T018	8,517	0.308	0.896	+0.065	reused

Main-run comparison and limits

The accepted main run C080_M020.T018 is the continuity baseline, not the top sensitivity setting. Across completed/reused temporary-CSI C/M/T runs, sensitivity changes magnitude more than direction. This sensitivity evidence is temporary-CSI only and does not support permanent-CSI sensitivity claims.

Incomplete grid disclosure

Three configurations remain blocked_partial: C080_M000.T012, C080_M000.T018, and C060_M020.T028.

Source: Sources: local AE-SENS summary CSVs and temporary transaction-cost overlay files.

Appendix A8: Limitations, Future Work, and Source Traceability

Limitations kept explicit

- Sensitivity evidence is temporary-CSI only.
- Test-set attribution caveats are documented where main-suite period=test rows are used.
- Model metrics do not establish index alpha by themselves.
- Feature-importance slides are directional perturbation evidence, not full AutoGluon-native importance.

Traceability pointer

- Dataset/model/index/sensitivity claims map to source files in the slide-source map and ticket evidence.
- Conceptual frames and future-work notes are marked as non-result claims.
- Future robustness: permanent-CSI sensitivity, low-vol/quality benchmarks, probability weighting, and terminal-failure miss decomposition.

Source-map status

This cleaned Draft appendix is a presentation-compression pass. The ticket evidence records the appendix remapping and the feature-importance source files used for A9–A10.

Appendix A9: Feature Importance – 11 Families

Temporary CSI, AG Expanded family-level perturbation; top canonical families

Family	Abs. re- sponse	Signed	N	Scale
Point-in-time ratios	0.609	-0.121	53	
Rolling-window statistics	0.523	-0.083	180	
Peak deterioration	0.152	-0.021	18	
Expanding volatility	0.098	+0.005	38	
YoY changes	0.094	-0.015	38	
Expanding mean	0.065	-0.007	38	
Trough rise	0.048	+0.002	7	
Price/macro interactions	0.039	-0.004	5	
Acceleration	0.037	-0.002	38	
Accounting momentum	0.037	-0.004	15	
Consecutive declines	0.008	+0.001	10	

Caveat

Model-based log-odds perturbation on bounded GBM-only predictor workspace; CV/training rows only; interpret as directional model-response evidence, not full AutoGluon-native feature importance.

Source: 03_Data_Output/10_FeatureImportance/family_importance/AE-FEAT-IMPORT-004R_family_importance_summary.csv.

Appendix A10: Feature Importance – Ratios and Individual Drivers

Point-in-time ratios

Ratio	Abs.	Signed
earn_yld	0.133	-0.017
altman_z2	0.131	+0.012
ocf_per_share	0.101	-0.015
roic	0.067	-0.019
roa	0.048	-0.020
ocf_margin	0.046	-0.009
interest_cov	0.042	-0.013
quick_ratio	0.028	-0.003

Individual features

Feature	Abs.	Family
unrate	0.181	PIT ratios
earn_yld	0.133	PIT ratios
altman_z2	0.131	PIT ratios
ocf_per_share	0.101	PIT ratios
peak_drop_log_mkvalt	0.096	Peak drop
roll_min_5y_earn_yld	0.086	Rolling
roll_min_3y_earn_yld	0.067	Rolling
roll_sd_5y_earn_yld	0.067	Rolling

Caveat

Model-based log-odds perturbation on bounded GBM-only predictor workspace; CV/training rows only; interpret as directional model-response evidence, not full AutoGluon-native feature importance.

Source: AE-FEAT-IMPORT-005R_pit_ratio_importance_summary.csv; AE-FEAT-IMPORT-006R_individual_feature_importance_summary.csv.

Impact of transaction costs: the winners are unchanged

Selected OOS winners when assuming transaction costs of 5, 10 and 20 bps.

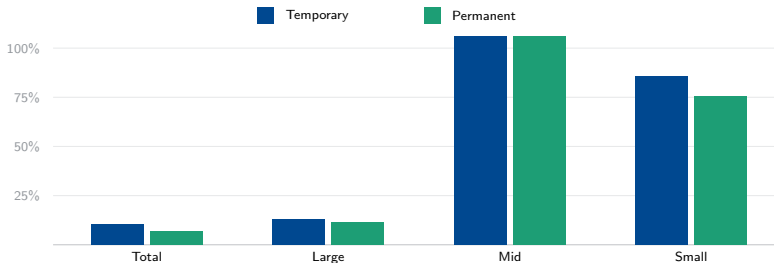
Track	Universe	Selected OOS winner	Active α (5/10/20)	Ranking
Temp.	Total	AG Base Dataset; Youden 3y	+0.43/ + 0.42/ + 0.41	unchanged
Temp.	Large	AG Base Dataset; Youden 3y	+0.15/ + 0.14/ + 0.12	unchanged
Temp.	Mid	AG Base Dataset; FPR5 5y	+0.45/ + 0.39/ + 0.28	unchanged
Temp.	Small	AG Exp. Dataset + VAE; Youden 3y	+0.59/ + 0.55/ + 0.45	unchanged
Perm.	Total	AG Exp. Dataset + VAE; FPR5	+0.26/ + 0.26/ + 0.25	unchanged
Perm.	Large	AG Exp. Dataset + VAE; FPR5	+0.22/ + 0.22/ + 0.20	unchanged
Perm.	Mid	AG Base Dataset; FPR5	+0.68/ + 0.62/ + 0.51	unchanged
Perm.	Small	AG Latent Dataset (VAE); FPR3	+0.28/ + 0.24/ + 0.16	unchanged

Takeaway

Raising costs to 20 bps lowers net returns but does not change a single winner or flip the strategy ranking.

Cost Drag due to turnover is concentrated in mid and small caps

Annualized gross buy + sell turnover of the selected OOS strategies (%)

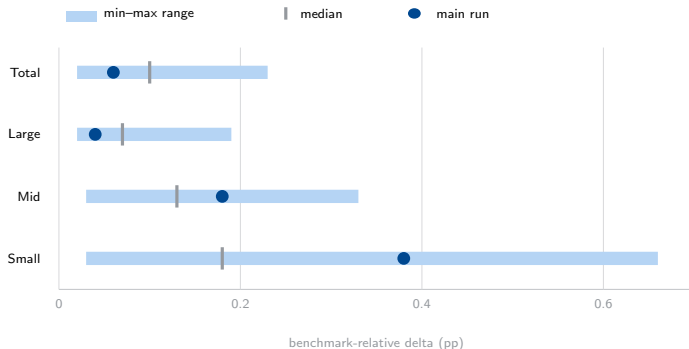


Summary

Drag scales linearly with turnover $\times$ cost. Total and Large are barely traded (between 7-13%), so their 20 bps drag is negligible. Mid and Small rebalancing is comparably larger at 75-106% a year, making their transaction-costs relevant for consideration. Yet, the winner ranking on the previous slide still holds.

Sensitivity Analysis: temporary CSI only

Main run versus 24 completed temporary-CSI C/M/T sensitivity runs; relative geometric-return delta (pp).



Robustness read

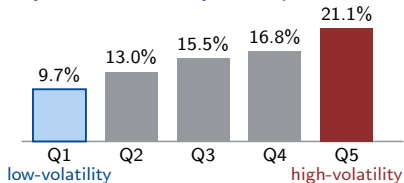
All completed temporary-CSI C/M/T runs stay positive; sensitivity changes magnitude, not sign. Main run: below median in Total/Large, above in Mid/Small. Temporary-CSI evidence only.

Same universe, window, and costs: what does the CSI screen add?

Methodology: low-volatility comparator

- Following the methodology by Baker et al. (2010), firms are sorted by trailing 5-year realized volatility into quintiles Q1 (low-vol)–Q5 (high-vol).
- The sort is rebuilt inside each benchmark universe: Total, Large, Mid, and Small are each comparable to the CSI-models.
- Portfolios are cap-weighted within quintile; low-vol and CSI rows are evaluated with 20 bps transaction costs, while the market benchmark remains the 0 bps reference.

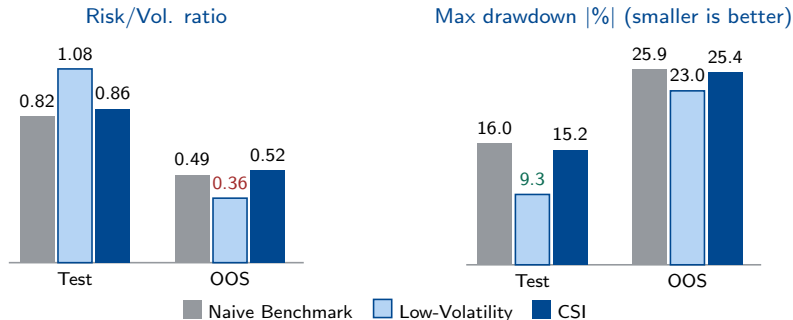
Realized volatility rises monotonically across quintiles within the test-set.



Motivation

The low-volatility screen removes firms because they are volatile. The CSI screen removes firms because the model ranks them as likely implosion candidates. The comparison asks whether that

Low-vol's edge is large, but conditional on the mark regime.



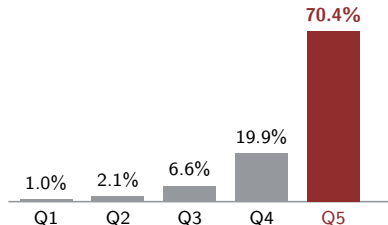
What it costs to be defensive (geo. return delta)

In the OOS-set, the Low-volatility portfolio performs considerably worse than in the test period (−3.4 pp). The CSI-model remains stable (+0.3 pp). The maximum drawdowns of the low-volatility portfolio are smaller than those of the benchmark and the CSI-model.

Remark: Avg. across four universes; 20 bps costs (Naive Benchmark 0 bps). CSI permanent tracks dynamic closely.

CSI-exclusions and the high-vol. portfolio overlap

Number of CSI-exclusions per decile portfolio



Reasons for the smaller return impact of CSI compared to low-vol:

Structural. CSI firms, dependent on the index, have a minor index weight. Their return impact is rather small

Statistical. Many FPs lead to a loss in performance due to their exclusion.

Summary

Low-vol performs well in the test-set, but it falls below the Benchmark in the OOS-set. CSI's gain is smaller but **regime-stable** and preserves the slight edge in returns and risk.

CSI complements low-vol. It does not dominate it.